

Contents

1	Lie Groups	2
1.1	Basic Definitions	2
1.2	Lie Subgroups	3
1.3	Group Actions and Equivariant Maps	3
1.4	Lie Bracket	5
1.5	The Lie Algebra of a Lie Group	6
1.6	One-Parameter Subgroups and the Exponential Map	10
	1.6.1 One-Parameter Subgroups	10
	1.6.2 The Exponential Map	11
1.7	The Closed Subgroup Theorem	13
1.8	The Lie Correspondence	13
1.9	Normal Subgroups	13
	1.9.1 The Adjoint Representation	14

Lie Groups

1.1 Basic Definitions

Definition 1.1.1

A **Lie group** is a smooth manifold G (without boundary) that is also a group in the algebraic sense, with the property that the multiplication map $m : G \times G \rightarrow G$ and inversion map $i : G \rightarrow G$, given by

$$m(g, h) = gh, \quad i(g) = g^{-1},$$

are both smooth.

Definition 1.1.2

If G is a Lie group, any element $g \in G$ defines maps $L_g, R_g : G \rightarrow G$, called **left translation** and **right translation**, respectively, by

$$L_g(h) = gh, \quad R_g(h) = hg.$$

which are diffeomorphisms.

Proof. Because L_g can be expressed as the composition of smooth maps

$$G \xrightarrow{\iota_g} G \times G \xrightarrow{m} G,$$

where $\iota_g(h) = (g, h)$ and m is multiplication, it follows that L_g is smooth. It is actually a diffeomorphism of G , because $L_{g^{-1}}$ is a smooth inverse for it. Similarly, $R_g : G \rightarrow G$ is a diffeomorphism. $\square$

1.2 Lie Subgroups

Definition 1.2.1

Suppose G is a Lie group. A **Lie subgroup of G** is a subgroup of G endowed with a topology and smooth structure making it into a Lie group and an **immersed submanifold** of G .

Remark.

- A Lie subgroup of a compact Lie group need not be compact.

Proposition 1.2.2

Suppose G is a Lie group, and $W \subseteq G$ is any neighborhood of the identity.

1. W generates an open subgroup of G .
2. If W is connected, it generates a connected open subgroup of G .
3. If G is connected, then W generates G .

Note.

S 生成子群的构造性表述, 就是 S 及 S^{-1} 的有限积的全体.

1.3 Group Actions and Equivariant Maps

Definition 1.3.1

- If G is a group and M is a set, a **left action of G on M** is a map $G \times M \rightarrow M$, often written as $(g, p) \mapsto g \cdot p$, that satisfies

$$\begin{aligned} g_1 \cdot (g_2 \cdot p) &= (g_1 g_2) \cdot p \quad \text{for all } g_1, g_2 \in G \text{ and } p \in M; \\ e \cdot p &= p \quad \text{for all } p \in M. \end{aligned} \tag{1.1}$$

- A **right action** is defined analogously as a map $M \times G \rightarrow M$ with the appropriate composition law:

$$\begin{aligned} (p \cdot g_1) \cdot g_2 &= p \cdot (g_1 g_2) \quad \text{for all } g_1, g_2 \in G \text{ and } p \in M; \\ p \cdot e &= p \quad \text{for all } p \in M. \end{aligned}$$

- If M is a topological space and G is a topological group, an action of G on M is said to be a **continuous action** if the defining map

$G \times M \rightarrow M$ or $M \times G \rightarrow M$ is continuous. In this case, M is said to be a **(left or right) G -space**.

- If in addition M is a smooth manifold with or without boundary, G is a Lie group, and the defining map is smooth, then the action is said to be a **smooth action**.

Definition 1.3.2

Sometimes it is useful to give a name to an action, such as $\theta : G \times M \rightarrow M$, with the action of a group element g on a point p usually written as $\theta_g(p)$.

- In terms of this notation, the conditions for a left action read

$$\theta_{g_1} \circ \theta_{g_2} = \theta_{g_1 g_2}, \quad \theta_e = \text{Id}_M, \quad (1.2)$$

- while for a right action the first equation is replaced by

$$\theta_{g_2} \circ \theta_{g_1} = \theta_{g_1 g_2}. \quad (1.3)$$

- For a smooth action, each map $\theta_g : M \rightarrow M$ is a diffeomorphism, because $\theta_{g^{-1}}$ is a smooth inverse for it.

Example 1.3.3

Every Lie Group G acts smoothly on itself by translation.

Proof. Given any two points $g_1, g_2 \in G$, there is a unique left translation of G taking g_1 to g_2 , namely left translation by $g_2 g_1^{-1}$; thus the action is both free and transitive. $\square$

Definition 1.3.4

Suppose $\theta : G \times M \rightarrow M$ is a left action of a group G on a set M .

- For each $p \in M$, the **orbit of p** , denoted by $G \cdot p$, is the set of all images of p under the action by elements of G :

$$G \cdot p = \{g \cdot p : g \in G\}.$$

- For each $p \in M$, the **isotropy group** or **stabilizer of p** , denoted by

G_p , is the set of elements of G that fix p :

$$G_p = \{g \in G : g \cdot p = p\}.$$

The definition of a group action guarantees that G_p is a subgroup of G .

- The action is said to be **transitive** if for every pair of points $p, q \in M$, there exists $g \in G$ such that $g \cdot p = q$, or equivalently if the only orbit is all of M .
- The action is said to be **free** if the only element of G that fixes any element of M is the identity: $g \cdot p = p$ for some $p \in M$ implies $g = e$, or equivalently if every isotropy group is trivial.

1.4 Lie Bracket

Definition 1.4.1 ► Lie Bracket

We define an operator $[X, Y] : C^\infty(M) \rightarrow C^\infty(M)$, called the **Lie bracket of X and Y** , by

$$[X, Y]f = XYf - YXf$$

Remark.

The Lie bracket of any pair of smooth vector fields is a smooth vector field.

Proposition 1.4.2 ► Coordinate Formula for the Lie Bracket

Let X, Y be smooth vector fields on a smooth manifold M with or without boundary, and let $X = X^i \frac{\partial}{\partial x^i}$ and $Y = Y^j \frac{\partial}{\partial x^j}$ be the coordinate expressions for X and Y in terms of some smooth local coordinates (x^i) for M . Then $[X, Y]$ has the following coordinate expression:

$$[X, Y] = \left(X^i \frac{\partial Y^j}{\partial x^i} - Y^i \frac{\partial X^j}{\partial x^i} \right) \frac{\partial}{\partial x^j}, \quad (1.4)$$

or more concisely,

$$[X, Y] = (XY^j - YX^j) \frac{\partial}{\partial x^j}. \quad (1.5)$$

Proposition 1.4.3 ► Naturality of the Lie Bracket

Let $F : M \rightarrow N$ be a smooth map between manifolds with or without boundary, and let $X_1, X_2 \in \mathfrak{X}(M)$ and $Y_1, Y_2 \in \mathfrak{X}(N)$ be vector fields such that X_i is F -related to Y_i for $i = 1, 2$. Then $[X_1, X_2]$ is F -related to $[Y_1, Y_2]$.

Proof. □

1.5 The Lie Algebra of a Lie Group

Definition 1.5.1

Suppose G is a Lie group. Recall that G acts smoothly and transitively on itself by left translation: $L_g(h) = gh$. A vector field X on G is said to be **left-invariant** if it is invariant under all left translations, in the sense that it is L_g -related to itself for every $g \in G$. More explicitly, this means

$$d(L_g)_{g'}(X_{g'}) = X_{gg'}, \quad \text{for all } g, g' \in G.$$

Since L_g is a diffeomorphism, this can be abbreviated by writing $(L_g)_*X = X$ for every $g \in G$.

Remark.

- Because $(L_g)_*(aX + bY) = a(L_g)_*X + b(L_g)_*Y$, the set of all smooth left-invariant vector fields on G is a linear subspace of $\mathfrak{X}(G)$.

Proposition 1.5.2

Let G be a Lie group, and suppose X and Y are smooth left-invariant vector fields on G . Then $[X, Y]$ is also left-invariant.

Proof. Let $g \in G$ be arbitrary. Since X, Y is L_g -related to itself. We have from the naturality of the Lie bracket that $[X, Y]$ is L_g -related to $[(L_g)_*X, (L_g)_*Y] = [X, Y]$. □

Corollary 1.5.3

The collection of all smooth left-invariant vector fields on a Lie group G is a Lie algebra under the Lie bracket, called the **Lie algebra of G** , and is denoted by $\text{Lie}(G)$.

Theorem 1.5.4

Let G be a Lie group. The evaluation map $\varepsilon: \text{Lie}(G) \rightarrow T_e G$, given by $\varepsilon(X) = X_e$, is a vector space isomorphism. Thus, $\text{Lie}(G)$ is finite-dimensional, with dimension equal to $\dim G$.

Proof sketch.

左不变向量场的全体是线性空间, ε 是线性映射.

按照左不变性将 $v \in T_e G$ 搬到全体 G 上, 得到一个唯一的左不变向量场.

Proof. It is clear from the definition that ε is linear over $\mathbb{R}$. It is easy to prove that it is injective: if $\varepsilon(X) = X_e = 0$ for some $X \in \text{Lie}(G)$, then the left-invariance of X implies that $X_g = d(L_g)_e(X_e) = 0$ for every $g \in G$, so $X = 0$.

To show that ε is surjective, let $v \in T_e G$ be arbitrary, and define a (rough) vector field v^L on G by

$$v^L|_g = d(L_g)_e(v)$$

If there is a left-invariant vector field on G whose value at the identity is v , clearly it has to be given this formula.

First we need to check that v^L is smooth. It suffices to show that $v^L f$ is smooth whenever $f \in C^\infty(G)$. Choose a $\square$

Proposition 1.5.5 ▶ Lie Algebra of the General Linear Group

The composition of the natural maps

$$\text{Lie}(\text{GL}(n, \mathbb{R})) \rightarrow T_{I_n} \text{GL}(n, \mathbb{R}) \rightarrow \mathfrak{gl}(n, \mathbb{R})$$

gives a Lie algebra isomorphism between $\text{Lie}(\text{GL}(n, \mathbb{R}))$ and the matrix algebra $\mathfrak{gl}(n, \mathbb{R})$.

Remark.

The vector spaces $\text{Lie}(\text{GL}(n, \mathbb{R}))$ and $\mathfrak{gl}(n, \mathbb{R})$ have independently de-

defined Lie algebra structures—the first coming from Lie brackets of vector fields, and the second from commutator brackets of matrices.

Proof sketch.

$T_{I_n} \text{GL}(n, \mathbb{R})$ 和 $\mathfrak{gl}(n, \mathbb{R})$ 有自然的同构, 前者有左不变向量场的李括号构成的李代数结构, 后者有矩阵的交换子构成的李代数结构.

$\text{Lie}(\text{GL}(n, \mathbb{R}))$ 有李括号构成的李代数结构, $\mathfrak{gl}(n, \mathbb{R})$ 有矩阵的交换子构成的李代数结构, 我们通过计算说明这两个结构是一致的.

左平移是 $\text{GL}(n, \mathbb{R})$ 上线性映射, 左不变向量场的微分是与其形式相同的线性映射.

Proof. Using the maxtrix entires X_j^i as the coordinates. There is a natural isomorphism

$$A_j^i \frac{\partial}{\partial X_j^i} \Big|_{I_n} \mapsto (A_j^i)$$

Then for each matrix $A = (A_j^i) \in \mathfrak{gl}(n, \mathbb{R})$, A gives a left-invariant vector field $A^L \in \mathfrak{g}$, by

$$A^L|_X = (dL_X) \left(A_j^i \frac{\partial}{\partial X_j^i} \right)$$

The differential of $L_X : A \mapsto XA$ is just the linear map of the same form , that is

$$A^L|_X = X_j^i A_k^j \frac{\partial}{\partial X_k^i}$$

Then

$$\begin{aligned} & [A^L|_X, B^L|_X] \\ &= \left(X_j^i A_k^j \frac{\partial}{\partial X_k^i} \right) \left(X_q^p B_r^q \frac{\partial}{\partial X_r^p} \right) - \left(X_q^p B_r^q \frac{\partial}{\partial X_r^p} \right) \left(X_j^i A_k^j \frac{\partial}{\partial X_k^i} \right) \\ &= X_j^i A_k^j B_r^k \frac{\partial}{\partial X_r^i} - X_q^p B_r^q A_k^r \frac{\partial}{\partial X_k^p} \\ &= X_j^i A_k^j B_r^k \frac{\partial}{\partial X_r^i} - X_i^j B_k^i A_r^k \frac{\partial}{\partial X_r^j} \\ &= \left(X_j^i A_k^j B_r^k - X_i^j B_k^i A_r^k \right) \frac{\partial}{\partial X_r^i} \end{aligned}$$

Evaluate at I_n , we get

$$[A^L|_X, B^L|_X]_{I_n} = (A_k^i B_r^k - B_k^i A_r^k) \frac{\partial}{\partial X_r^i} = (AB - BA)_r^i \frac{\partial}{\partial X_r^i}$$

□

Induced Lie Algebra Homomorphisms

Theorem 1.5.6 ▶ Induced Lie Algebra Homomorphisms

Let G and H be Lie groups, and let $\mathfrak{g}$ and $\mathfrak{h}$ be their Lie algebras. Suppose $F : G \rightarrow H$ is a Lie group homomorphism. For every $X \in \mathfrak{g}$, there is a unique vector field in $\mathfrak{h}$ that is F -related to X . With this vector field denoted by F_*X , the map $F_* : \mathfrak{g} \rightarrow \mathfrak{h}$ so defined is a Lie algebra homomorphism, is called the **induced Lie algebra homomorphism**.

Remark.

Note that the theorem implies that for any left-invariant vector field $X \in \mathfrak{g}$, F_*X is a well-defined smooth vector field on H , even though F may not be a diffeomorphism.

Proof sketch.

- 左不变性使得相关向量场的选取是唯一且确定的, 唯一需要验证的是相关性.
- 群同态保证了关于平移的自然性, $F \circ L_g = L_{F(g)} \circ F$. 它的微分形式 $dF \circ d(L_g) = d(L_{F(g)}) \circ dF$. 既是左不变向量场之间的相关性.
- F_* 的 Lie 代数同态性, 来自于李括号的相关性 $F_*[X, Y] = [F_*X, F_*Y]$.

The Lie Algebra of the Lie Subgroup

1.6 One-Parameter Subgroups and the Exponential Map

1.6.1 One-Parameter Subgroups

Definition 1.6.1

A **one-parameter subgroup** of G is defined to be a Lie group homomorphism $\gamma : \mathbb{R} \rightarrow G$, with $\mathbb{R}$ considered as a Lie group under addition.

Theorem 1.6.2 ► Characterization of One-Parameter Subgroups

Let G be a Lie group. The one-parameter subgroups of G are precisely the maximal integral curves of left-invariant vector fields starting at the identity.

Remark.

Given $X \in \text{Lie}(G)$, the one-parameter subgroup determined by X in this way is called the **one-parameter subgroup generated by X** . Because left-invariant vector fields are uniquely determined by their values at the identity, it follows that each one-parameter subgroup is uniquely determined by its initial velocity in $T_e G$, and thus there are one-to-one correspondences

$$\{\text{one-parameter subgroups of } G\} \longleftrightarrow \text{Lie}(G) \longleftrightarrow T_e G.$$

Proof sketch.

$\Leftarrow$: 给定积分曲线 γ , 为了证明群律 $\gamma(s+t) = \gamma(s)\gamma(t)$, 以重参数化和左平移两种方式将积分曲线的起点搬到 $\gamma(s)$, 其中后者依赖左不变向量场的平移不变形.

$\Rightarrow$: 群同态 γ 给出李代数同态 γ_* , $\gamma_*\left(\frac{d}{dt}\right)$ 就是一个左不变向量场, 它恰好是 γ 的切向量场.

Proof. Supposet that γ is a maximal integral curve of a left-invariant vector field X . Since X is complete, γ has the domain $\mathbb{R}$. The left-invariance means that X is L_g -invariant for each $g \in G$, then L_g take the integral curve of X to the integral curve of X . $t \mapsto L_{\gamma(s)} \circ \gamma(t)$ is the integral curve of X starting at $\gamma(s)$. The rescaling lemma shows that $t \mapsto \gamma(s+t)$ is also the integral curve

of X starting at $\gamma(s)$, then it follows from the uniqueness of the integral curve that

$$\gamma(s+t) = L_{\gamma(s)} \circ \gamma(t) = \gamma(s) \gamma(t)$$

Conversely, if γ is the one-parameter subgroup of G . Let $X = \gamma_* \left(\frac{d}{dt} \right) \in \text{Lie}(G)$, treating $\frac{d}{dt}$ as a left-invariant vector field on $\mathbb{R}$. $\square$

Proposition 1.6.3

Suppose G is a Lie group and $H \subseteq G$ is a Lie subgroup. The one-parameter subgroups of H are precisely those one-parameter subgroups of G whose initial velocities lie in $T_e H$.

Proof sketch.

单参数子群与切空间是对应的。

单参数子群就是过原点的积分曲线，只要初速度一样，它们就是相同的积分曲线。

1.6.2 The Exponential Map

Definition 1.6.4 ▶ Exponential map

Given a Lie group G with Lie algebra $\mathfrak{g}$, we define a map $\exp : \mathfrak{g} \rightarrow G$, called the **exponential map of G** , as follows: for any $X \in \mathfrak{g}$, we set

$$\exp X = \gamma(1),$$

where γ is the one-parameter subgroup generated by X , or equivalently the integral curve of X starting at the identity.

Proposition 1.6.5 ▶ Properties of the Exponential Map

Let G be a Lie group and let $\mathfrak{g}$ be its Lie algebra.

- (a) The exponential map is a smooth map from $\mathfrak{g}$ to G .
- (b) For any $X \in \mathfrak{g}$ and $s, t \in \mathbb{R}$, $\exp(s+t)X = \exp sX \exp tX$.
- (c) For any $X \in \mathfrak{g}$, $(\exp X)^{-1} = \exp(-X)$.
- (d) For any $X \in \mathfrak{g}$ and $n \in \mathbb{Z}$, $(\exp X)^n = \exp(nX)$.
- (e) The differential $(d \exp)_0 : T_0 \mathfrak{g} \rightarrow T_e G$ is the identity map, under the canonical identifications of both $T_0 \mathfrak{g}$ and $T_e G$ with $\mathfrak{g}$ itself.
- (f) The exponential map restricts to a diffeomorphism from some

neighborhood of 0 in $\mathfrak{g}$ to a neighborhood of e in G .

- (g) If H is another Lie group, $\mathfrak{h}$ is its Lie algebra, and $\Phi : G \rightarrow H$ is a Lie group homomorphism, the following diagram commutes:

$$\begin{array}{ccc} \mathfrak{g} & \xrightarrow{\Phi_*} & \mathfrak{h} \\ \exp \downarrow & & \downarrow \exp \\ G & \xrightarrow{\Phi} & H. \end{array} \quad (1.6)$$

- (h) The flow θ of a left-invariant vector field X is given by $\theta_t = R_{\exp tX}$ (right multiplication by $\exp tX$).

Proposition 1.6.6 ►

Let G be a Lie group. For any $X \in \text{Lie}(G)$, $\gamma(s) = \exp sX$ is the one-parameter subgroup of G generated by X .

Proof. Let $\gamma : \mathbb{R} \rightarrow G$ be the one-parameter subgroup generated by X , which is the integral curve of X starting at e . For any fixed $s \in \mathbb{R}$, the rescaling lemma of integral curve gives that $\tilde{\gamma}(t) = \gamma(st)$ is the integral curve of sX starting at e , so

$$\exp sX = \tilde{\gamma}(1) = \gamma(s)$$

□

Proposition 1.6.7

Let G be a Lie group, and let $H \subseteq G$ be a Lie subgroup. With $\text{Lie}(H)$ considered as a subalgebra of $\text{Lie}(G)$ in the usual way, the exponential map of H is the restriction to $\text{Lie}(H)$ of the exponential map of G , and

$$\text{Lie}(H) = \{X \in \text{Lie}(G) : \exp tX \in H \text{ for all } t \in \mathbb{R}\}.$$

1.7 The Closed Subgroup Theorem

Theorem 1.7.1 ► Closed Subgroup Theorem

Suppose G is a Lie group and $H \subseteq G$ is a subgroup that is also a closed subset of G . Then H is an embedded Lie subgroup.

1.8 The Lie Correspondence

Theorem 1.8.1 ►

Suppose G and H are Lie groups with G simply connected, and let $\mathfrak{g}$ and $\mathfrak{h}$ be their Lie algebras. For any Lie algebra homomorphism $\varphi : \mathfrak{g} \rightarrow \mathfrak{h}$, there is a unique Lie group homomorphism $\Phi : G \rightarrow H$ such that $\Phi_* = \varphi$.

Corollary 1.8.2

If G and H are simply connected Lie groups with isomorphic Lie algebras, then G and H are isomorphic.

Theorem 1.8.3 ► The Lie Correspondence

There is a one-to-one correspondence between isomorphism classes of finite-dimensional Lie algebras and isomorphism classes of simply connected Lie groups, given by associating each simply connected Lie group with its Lie algebra.

1.9 Normal Subgroups

Lemma 1.9.1

Let G be a connected Lie group, and let $H \subseteq G$ be a connected Lie subgroup. Let $\mathfrak{g}$ and $\mathfrak{h}$ denote the Lie algebras of G and H , respectively. Then H is normal in G if and only if

$$(\exp X)(\exp Y)(\exp(-X)) \in H \quad \text{for all } X \in \mathfrak{g} \text{ and } Y \in \mathfrak{h}. \quad (1.7)$$

Proof sketch.

$\exp$ 是 $0 \in \mathfrak{h}$ 到 $e \in H$ 的局部的微分同胚, 也是 $0 \in \mathfrak{g}$ 到 $e \in G$ 的微分同胚. 于是 $e \in H$ 和 $e \in G$ 分别有一个以 $\exp$ 像构成的开邻域 U, U_0 . 而连通李群的单位元的开邻域生成整个李群, 故 U 和 U_0 分别生成了 H 和 G .

题设条件应用在这两个 $\exp$ 像构成的开邻域上, 给出 U 中元素在 U_0 从元素的共轭下落在 H 中. 利用生成的性质 (H 和 G 中的元素分别写成 U 和 U_0 中的元素的有限积). 我们可以将 ghg^{-1} 拆分成若干 $g_i h_j g_i^{-1}$ 的组合, 其中 $g_i \in U_0, h_j \in U$.

1.9.1 The Adjoint Representation

Definition 1.9.2

Let G be a Lie group and $\mathfrak{g}$ be its Lie algebra. For any $g \in G$, the conjugation map $C_g : G \rightarrow G$ given by $C_g(h) = ghg^{-1}$ is a Lie group homomorphism. We let $\text{Ad}(g) = (C_g)_* : \mathfrak{g} \rightarrow \mathfrak{g}$ denote its induced Lie algebra homomorphism.

Proposition 1.9.3 ▶ The Adjoint Representation

If G is a Lie group with Lie algebra $\mathfrak{g}$, the map $\text{Ad} : G \rightarrow \text{GL}(\mathfrak{g})$ is a Lie group representation, called the *adjoint representation* of G .

Definition 1.9.4

Given a finite dimensional Lie algebra $\mathfrak{g}$, for each $X \in \mathfrak{g}$, define a map $\text{ad}(X) : \mathfrak{g} \rightarrow \mathfrak{g}$ by $\text{ad}(X)Y = [X, Y]$.

Proposition 1.9.5 ▶ adjoint representation of $\mathfrak{g}$

For any Lie algebra $\mathfrak{g}$, the map $\text{ad} : \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$ is a Lie algebra representation, called the *adjoint representation of $\mathfrak{g}$* .

Theorem 1.9.6

Let G be a Lie group, let $\mathfrak{g}$ be its Lie algebra, and let $\text{Ad} : G \rightarrow \text{GL}(\mathfrak{g})$ be the adjoint representation of G . The induced Lie algebra representation $\text{Ad}_* : \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$ is given by $\text{Ad}_* = \text{ad}$.